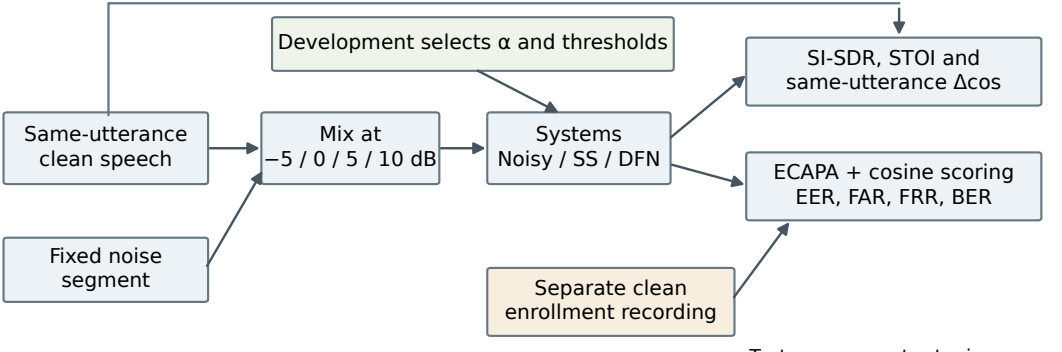
Development selects α and thresholds
Same-utterance clean speech
Mix at −5 / 0 / 5 / 10 dB
Systems Noisy / SS / DFN
SI-SDR, STOI and same-utterance Δcos
ECAPA + cosine scoring EER, FAR, FRR, BER
Fixed noise segment
Separate clean enrollment recording
Test: no parameter tuning